21. ROS 2 URDF Model

1. Introduction to URDF

The modeling method in ROS is called URDF, whose full name is Unified Robot Description Format. It can clearly describe not only the robot's own model but also the robot's external environment. URDF model files use the XML format.

SolidWorks model to URDF plugin: [sw_urdf_exporter - ROS Wiki]

2. Robot Components

When modeling and describing a robot, you first need to be familiar with the robot's components and parameters. For example, a robot is generally composed of four major parts: **hardware structure, drive system, sensor system, and control system**. Common robots on the market, whether mobile robots or robotic arms, can be decomposed according to these four components.



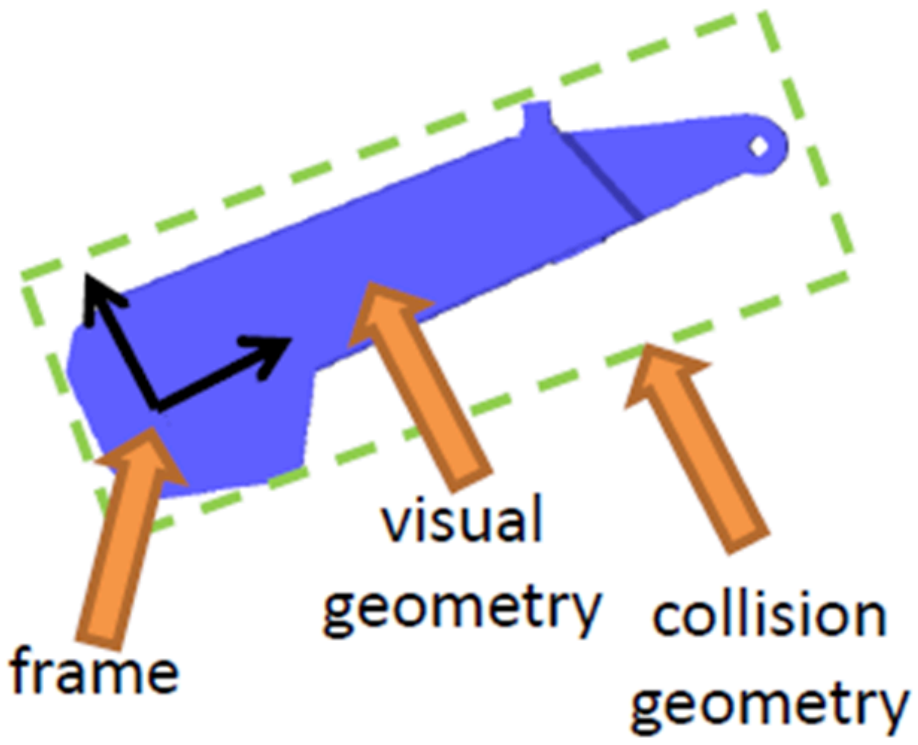
- The hardware structure refers to the devices that can actually be seen, such as the chassis, housing, and motors;
- The drive system refers to the devices that enable these devices to operate normally, such as the motor driver and the power management system;
- The sensor system includes the encoder on the motor, the onboard IMU, the installed camera, radar, and so on, so that the robot can perceive its own state and the external environment;
- The control system is the main carrier of our development process, generally a computing platform such as a Raspberry Pi or a computer, together with its operating system and application software.

The robot modeling process actually follows a similar approach: describe every part of the robot clearly with a modeling language, and then combine them.

3. URDF Syntax

3.1. Link Description

The tag is used to describe the appearance and physical properties of a rigid body part of the robot. The appearance includes size, color, and shape, while the physical properties include mass, inertia matrix, collision parameters, and so on.



Taking this robotic arm link as an example, its link description is as follows:

```
<link name="link_4">
    <visual>
        <geometry>
            <mesh filename="link_4.stl"/>
        </geometry>
        <origin xyz="0 0 0" rpy="0 0 0" />
    </visual>
    <collision>
        <geometry>
            <cylinder length="0.5" radius="0.1"/>
        </geometry>
        <origin xyz="0 0 -0.05" rpy="0 0 0" />
    </collision>
</link>
```

The name in the link tag indicates the name of the link. You can customize it, and this name will be used when a joint connects links later.

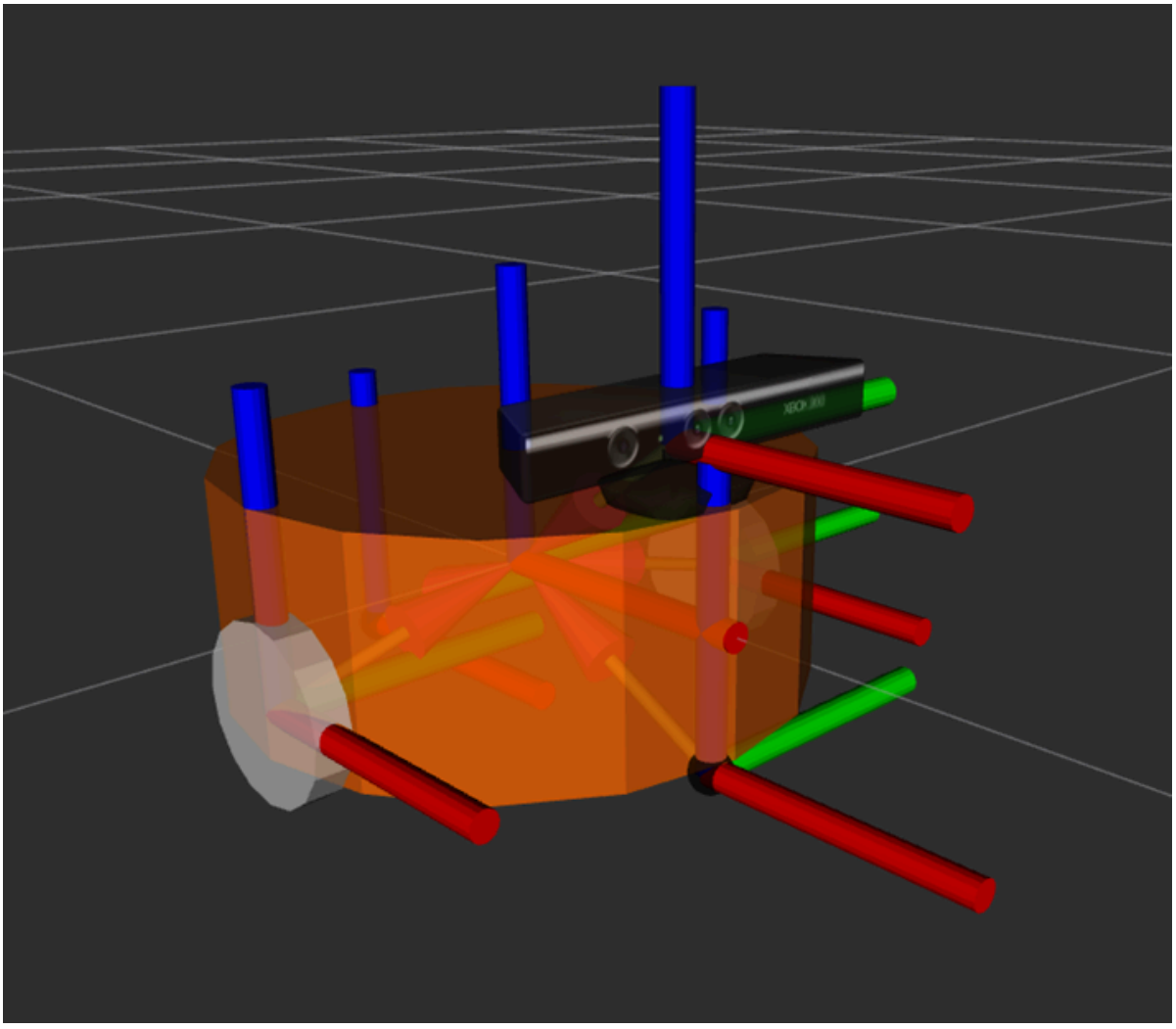
The part inside the link is used to describe the robot's appearance, for example:

- indicates the geometric shape. Inside it, references a blue appearance designed in advance in 3D software, that is, this STL file, which looks consistent with the real robot.
- indicates the offset of the coordinate frame relative to the initial position, namely the translation along the x, y, and z directions and the roll, pitch, and raw rotations. If no offset is required, all of them are 0.

The second part, , describes the collision parameters. Its content seems to be the same as , since it also has and . They look alike, but the difference is actually quite large.

- The part focuses on describing how the robot looks, that is, the visual effect;
- The part describes the robot's state during motion, for example, how contact between the robot and the outside world counts as a collision.

In this robot model, the blue part is described by . In actual control, such a complex appearance requires high computing power when collision detection is calculated. To simplify the calculation, we simplified the model used for collision detection to the cylinder in the green box, that is, the shape described by in . The coordinate frame offset is similar and can describe the offset of the rigid body's center of mass.



For a mobile robot, link can also be used to describe parts of the car such as the body and the wheels.

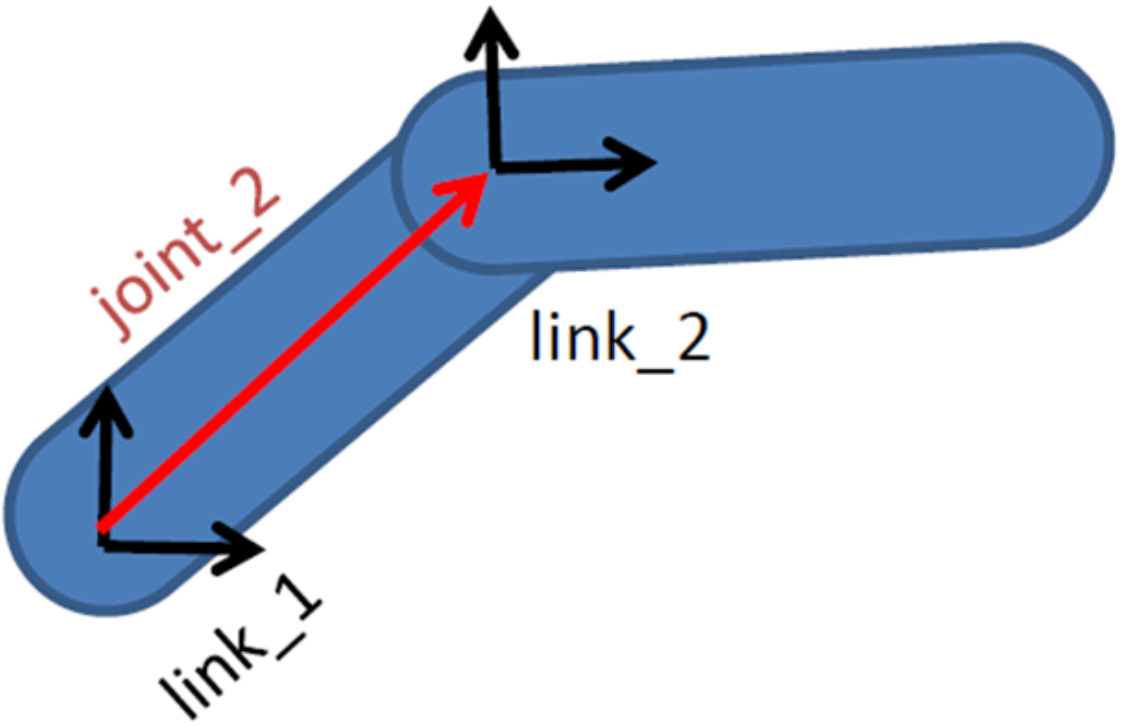
3.2. Joint Description

Rigid bodies in a robot model can only produce relative motion after they are connected through joints.

Joints in URDF have six motion types.

Joint Type	Description
continuous	Revolute joint that can rotate infinitely about a single axis
revolute	Revolute joint, similar to continuous, but with angular limits on rotation
prismatic	Prismatic joint that slides along an axis, with position limits
fixed	Fixed joint, a special joint that does not allow motion
floating	Floating joint that allows translation and rotation
planar	Planar joint that allows translation or rotation in directions orthogonal to the plane

1. continuous, which describes rotational motion and can rotate infinitely around one axis, such as the wheels of the car.
2. revolute, which is also a revolute joint. The difference from the continuous type is that it cannot rotate infinitely but has an angle limit, such as the two links of a robotic arm.
3. prismatic, which is a sliding joint and can translate along one axis, also with a position limit. A linear motor generally uses this motion mode.
4. fixed, a fixed joint, which is the only joint that does not allow motion. However, it is used quite frequently. For example, the relative position of a camera link installed on the robot does not change, and the connection mode used at this time is Fixed.
5. Floating is a floating joint, and the sixth type, planar, is a planar joint. These two are used relatively rarely.

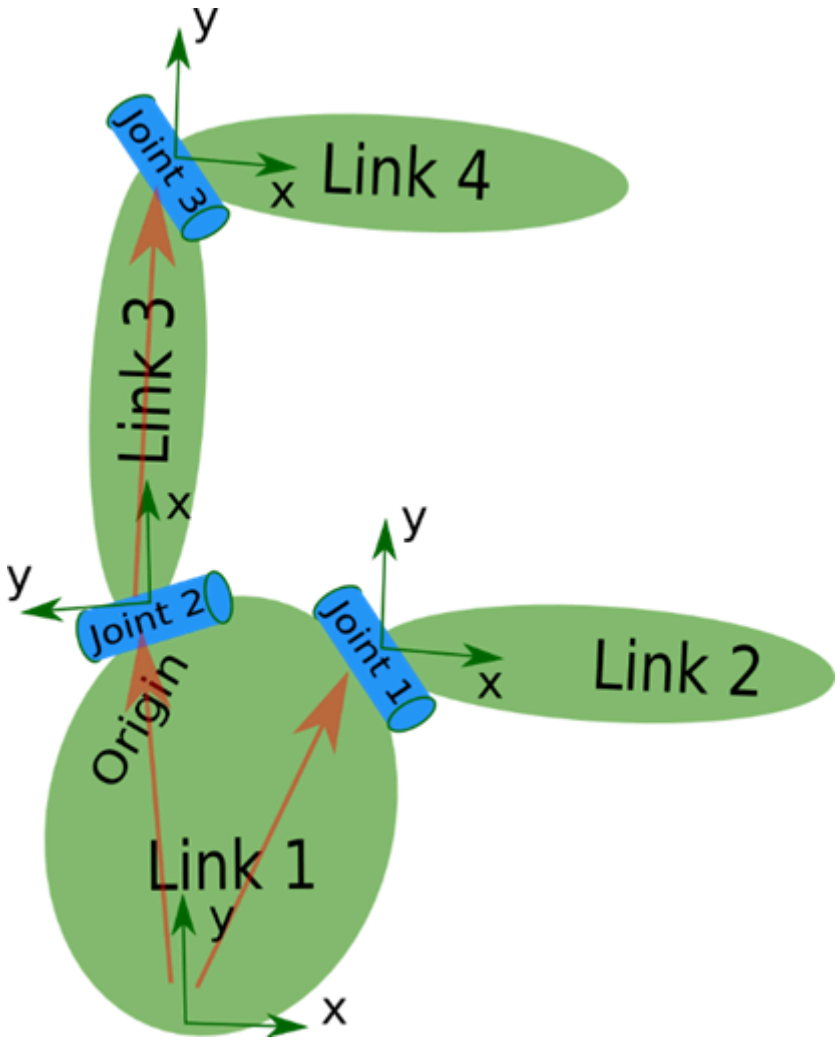


In a URDF model, every link is described with such a piece of XML content, for example, the name of the joint and the motion type.

```
<joint name="joint_2" type="revolute">
    <parent link="link_1"/>
    <child link="link_2"/>
    <origin xyz="0.2 0.2 0" rpy="0 0 0"/>
    <axis xyz="0 0 1"/>
    <limit lower="-3.14" upper="3.14" velocity="1.0"/>
</joint>
```

- parent tag: describes the parent link;
- child tag: describes the child link, and the child link moves relative to the parent link;
- origin: indicates the relationship between the coordinate frames of the two links, that is, the red vector in the figure. It can be understood as how the two links should be mounted together;
- axis indicates the unit vector of the joint motion axis. For example, if z equals 1, it means the rotational motion is around the positive direction of the z-axis;
- limit indicates some limits on the motion, such as the minimum position, the maximum position, and the maximum velocity.

4. Complete Robot Model



Finally, all the link and joint tags complete the description and combination of every part of the robot. All of them are placed in one robot tag, forming the complete robot model.

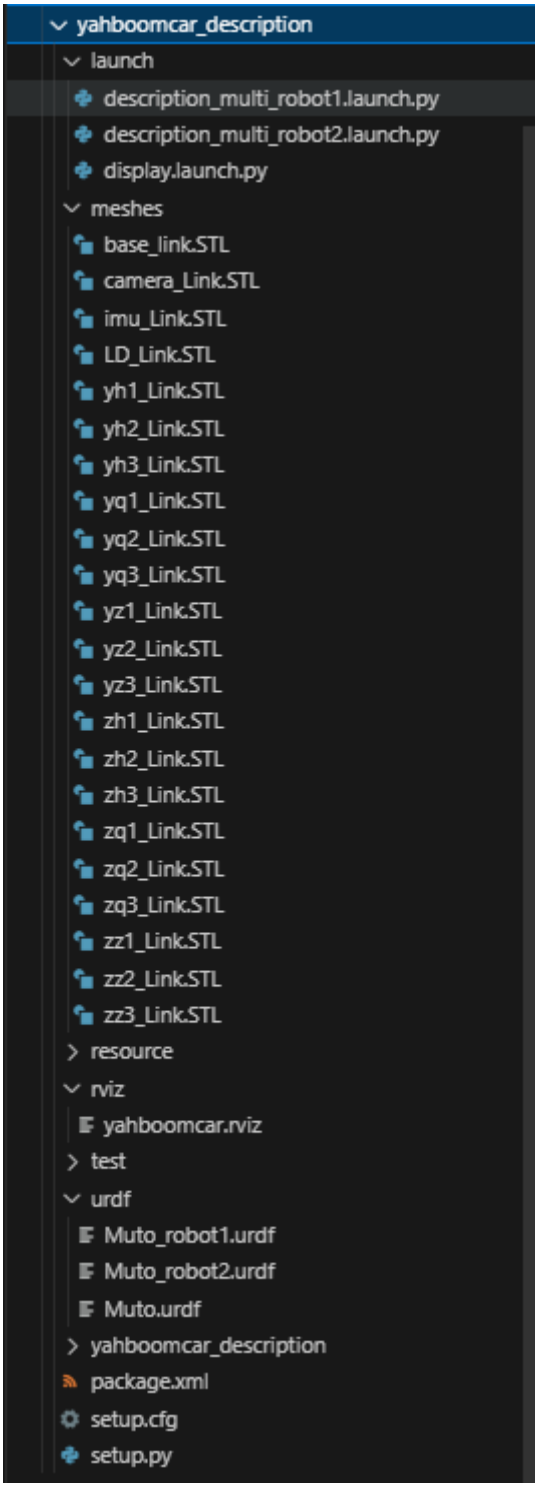
```
<robot name="<name of the robot>">
     <link> ..... </link>
     <link> ..... </link>

     <joint> ..... </joint>
     <joint> ..... </joint>
</robot>
```

Therefore, when looking at a URDF model, do not rush to read the details of each piece of code. First find the links and joints to see which parts the robot consists of, and after understanding the whole picture, look at the details.

4.1. Creating a Robot Model

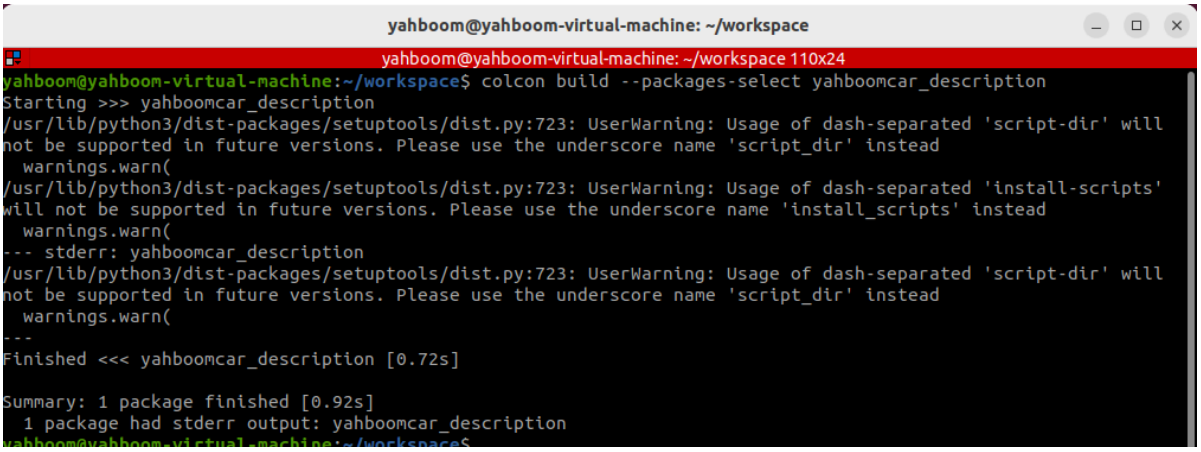
Taking the muto model as an example, copy the `yahboomcar_description` package in this lesson's folder to the src directory of the workspace:



- urdf: stores the URDF or xacro files of the robot model
- meshes: stores the model rendering files referenced in the URDF
- launch: stores the related launch files
- rviz: stores the rviz configuration file

Then build the package.

```
colcon build --packages-select yahboomcar_description
```



4.2. Model Visualization Effect

- Refresh the environment variables, and then run the launch command.

```
ros2 launch yahboomcar_description display.launch.py
```

After that, rviz starts automatically and displays the robot model:

